

MATH 2531 Chapter 15 Multiple Integrals

A complete problem set

Draw every region, shadow, and solid by hand before setting up an integral. Within each section the problems run from picture first and conceptual to a final computation at the level of the exam, medium hard for an average student. Some problems draw on vectors, surfaces, and gradients from Chapters 12 to 14. No Lagrange multipliers are needed anywhere.

1 Seeing the Volume (draw first)

1. [1D] Sketch $y = 1 - x^2$ on $[-1, 1]$ and shade the area $\int_{-1}^1 (1 - x^2) dx$. This is one vertical slice of the dome $z = 1 - x^2 - y^2$.
2. [2D IN 3D] Sketch the dome $z = 1 - x^2 - y^2$ over the unit disk, and beside it sketch its shadow on the xy -plane. Which region sets the limits of the double integral for the volume?
3. [2D IN 3D] On the dome picture draw a thin slab in the y -direction at a fixed x . Give its lower and upper limits in y and the height it carries, then write $\iint_D (1 - x^2 - y^2) dA$ as an iterated integral.
4. Draw the solid under $z = 4 - x - y$ over the square $[0, 1] \times [0, 1]$ and label the four corner heights. Estimate the volume from the corner heights.
5. Draw the region for $\int_0^1 \int_{x^2}^x f(x, y) dy dx$, shade a vertical strip, then redraw it and shade a horizontal strip for the reversed order.
6. [2D IN 3D] Draw the paraboloid $z = x^2 + y^2$ and the plane $z = 4$. Shade the solid between them and draw its shadow. What curve is the shadow's boundary?

2 Double Integrals over Rectangles

1. [2D] For $R = [0, 4] \times [-1, 2]$ estimate $\iint_R (1 - xy^2) dA$ with a Riemann sum, $m = 2$, $n = 3$, sampling the lower right corner of each subrectangle. Sketch the grid and mark the sample points.
2. Use the Midpoint Rule with $m = n = 2$ to estimate the volume under $z = 1 + x^2 + 3y$ over $R = [1, 2] \times [0, 3]$.
3. A contour map of f on $[0, 4] \times [0, 4]$ has evenly spaced level curves. Sketch a plausible map, then use the Midpoint Rule with $m = n = 2$ to estimate $\iint_R f dA$ and the average value of f .
4. Evaluate $\iint_R \sqrt{7} dA$ over $R = [2, 6] \times [-1, 5]$ by reading it as the volume of a box.
5. Evaluate $\int_0^2 \int_0^3 (x + 2y) dy dx$, then redo it in the order $dx dy$ and confirm Fubini gives the same value.

- Evaluate $\iint_R xy \, dA$ over $R = [0, 1] \times [0, 2]$, and show it factors as a product of two single integrals. When does a double integral separate this way?
- Find the volume of the solid under $z = 4 - x - y$ over $R = [0, 1] \times [0, 2]$.
- Find the average value of $f(x, y) = x^2y$ over $R = [0, 2] \times [0, 1]$.

3 Double Integrals over General Regions

- [2D] Sketch the region D bounded by $y = x$ and $y = x^2$. Draw a vertical slice of its area and give the y limits it sweeps, then draw a horizontal slice and give the x limits it sweeps.
- Sketch the region bounded by $y = x - 2$, $y = \sqrt{x}$, and $y = 6 - x$, and set up $\iint_D (x + y) \, dA$ in the easier order.
- Find the area of the region bounded by $y = x$ and $y = x^2 - 2x$ using $\iint_D 1 \, dA$, and draw the region.
- [2D IN 3D] The paraboloid $z = x^2 + y^2$ meets the plane $z = 2x$ in a curve. Show its shadow is the disk $(x - 1)^2 + y^2 \leq 1$, sketch that disk, and set up the volume between the two surfaces over it.
- Find the volume under $z = 1 + x + y$ and above the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$.
- Find the volume under the paraboloid $z = x^2 + y^2$ over the region D bounded by $y = x^2$ and $x = y^2$. Draw D .
- Evaluate $\iint_D (x + y) \, dA$ over the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$.
- Evaluate $\iint_D x \, dA$ over the region bounded by $y = x$ and $y = x^2$.

4 Reversing the Order of Integration

- [2D] Sketch the region and reverse the order for $\int_0^1 \int_0^{x^2} f(x, y) \, dy \, dx$.
- Explain in one sentence, using a region sketch, why reversing the order sometimes turns an impossible antiderivative into an easy one.
- Reverse the order and evaluate $\int_0^1 \int_{3y}^3 e^{x^2} \, dx \, dy$.
- Evaluate $\int_0^1 \int_y^1 \frac{\sin x}{x} \, dx \, dy$ by reversing the order.
- Evaluate $\int_0^1 \int_{x^2}^1 x^3 \sin(y^3) \, dy \, dx$.
- Evaluate $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$ by reversing the order. Draw the region.

5 Double Integrals in Polar Coordinates

1. [2D] Sketch each region and write $\iint_R f \, dA$ in polar, an upper half disk of radius 2, a quarter disk in the first quadrant of radius 3, and the annulus $1 \leq r \leq 2$.
2. Sketch the region and evaluate $\int_{\pi/4}^{3\pi/4} \int_1^2 r \, dr \, d\theta$.
3. Sketch $r = 2 \sin \theta$, identify it as a circle, and evaluate $\int_0^{\pi/2} \int_0^{2 \sin \theta} r \, dr \, d\theta$.
4. Sketch the rose $r = \cos 3\theta$ and find the area of one loop, then sketch the cardioid $r = 1 + \cos \theta$ and find the area it encloses.
5. Evaluate $\iint_D (x^2 + y^2) \, dA$ over the disk $x^2 + y^2 \leq 4$.
6. Evaluate $\iint_D e^{-(x^2+y^2)} \, dA$ over the disk of radius 2 centered at the origin.
7. Find the volume of the solid below the cone $z = \sqrt{x^2 + y^2}$ and above the disk $x^2 + y^2 \leq 4$. Draw the solid.
8. Convert to polar and evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$. Draw the region and mark why polar is natural.
9. Use polar coordinates to find the volume under $z = 4 - x^2 - y^2$ and above the xy -plane. Draw the solid and its shadow.

6 Applications of Double Integrals

1. [2D] Sketch the lamina bounded by $y = \sqrt{x}$, $y = 0$, $x = 1$ with density $\rho(x, y) = x$, shading darker where it is heavier. Find its mass.
2. Electric charge on the disk $x^2 + y^2 \leq 1$ has density $\sigma(x, y) = \sqrt{x^2 + y^2}$. Find the total charge.
3. The temperature on the plate $[0, 2] \times [0, 1]$ is $T(x, y) = x^2 + y$. Find the average temperature.
4. Find the centroid of the region bounded by $y = x$ and $y = x^2$, and mark it on the region.
5. Find the mass and center of mass of the rectangle $[0, 2] \times [0, 1]$ with density $\rho(x, y) = 1 + x$.
6. [APPLICATION] A joint probability density on the unit square is $f(x, y) = x + y$. Verify $\iint f \, dA = 1$, find $P(X \leq \frac{1}{2}, Y \leq \frac{1}{2})$ and shade that region, then find the marginal density $f_X(x)$ and the expected value $E[X] = \iint x f \, dA$.
7. Let $f(x, y) = c e^{-(x+y)}$ on the first quadrant. Find c so the total is one, draw the region $X + Y \leq 1$, and find $P(X + Y \leq 1)$.
8. Find the mass and center of mass of the triangle with vertices $(0, 0)$, $(2, 0)$, $(0, 2)$ with density $\rho(x, y) = x + y$, and mark the balance point.

7 Triple Integrals over Boxes and General Solids

1. [3D] Evaluate $\iiint_E xyz \, dV$ over the box $[0, 1] \times [0, 2] \times [0, 3]$.
2. Sketch the solid whose volume is $\int_0^2 \int_0^{2-y} \int_0^{4-y^2} dx \, dz \, dy$ and describe its bounding surfaces.
3. Find the volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$. Draw the solid and its shadow.
4. Evaluate $\iiint_T y^2 \, dV$ over the tetrahedron with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$.
5. Find the average value of $f(x, y, z) = x + y + z$ over the unit cube $[0, 1]^3$.
6. Find the mass of the solid bounded above by $z = 4 - x^2 - y^2$ and below by the xy -plane with density $\delta = z$. Draw the solid.
7. [2D IN 3D] The plane $2x + y + z = 4$ crosses the axes at three points. Draw the tetrahedron it cuts in the first octant with the coordinate planes, then find its volume.
8. Consider $\int_0^1 \int_0^{1-x} \int_0^2 dz \, dy \, dx$. Sketch the solid E it describes, then rewrite the integral in the order $dx \, dy \, dz$.

8 Triple Integrals in Cylindrical Coordinates

1. [3D] Sketch the solid inside $x^2 + y^2 = 1$, below $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$, and mark its shadow.
2. Sketch the solid $r^2 \leq z \leq 8 - r^2$ and set up its volume in cylindrical coordinates.
3. Find the mass of the cylinder $x^2 + y^2 \leq 1$, $0 \leq z \leq 2$, with density $\rho = z$.
4. Evaluate $\iiint_E \sqrt{x^2 + y^2} \, dV$ inside the cylinder $x^2 + y^2 \leq 4$ between $z = 0$ and $z = 3$.
5. Convert to cylindrical and evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \int_{-2}^2 xz \, dz \, dx \, dy$.
6. Evaluate $\iiint_E z \, dV$ over the solid enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
7. Use cylindrical coordinates to find the volume of the solid bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 4$. Sketch the solid and mark its shadow.

9 Triple Integrals in Spherical Coordinates

1. [3D] Draw a single spherical wedge and label the three edge lengths $d\rho$, $\rho \, d\phi$, and $\rho \sin \phi \, d\theta$ whose product gives $\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.
2. Find the volume of a ball of radius a using spherical coordinates.

3. Find the mass of the ball $\rho \leq 1$ with density $\delta = \rho$.
4. Evaluate $\iiint_B (x^2 + y^2 + z^2) dV$ over the ball $\rho \leq 2$.
5. Evaluate $\iiint_E z dV$ over the first octant part of the region between the spheres $\rho = 1$ and $\rho = 3$.
6. Evaluate $\iiint_E z dV$ over the upper half of the ball $\rho \leq 1$ with $z \geq 0$.
7. Find the volume of the solid above the cone $z = \sqrt{x^2 + y^2}$ and inside the sphere $x^2 + y^2 + z^2 = 2$. Draw it.
8. Find the volume of the ice cream cone that lies above the cone $\phi = \pi/3$ and inside the sphere $\rho = 2$. Sketch it.